

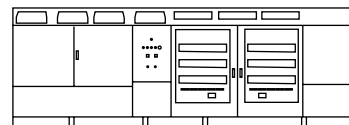


PCSM

IEC

Easy maintenance.
Integrated MV solution in the same enclosure.
Advanced grid support.
Compatible with all battery technologies.





REFERENCES	FP5150MH	FP5151MH	FP5152MH	
AC	AC Output Power (kVA/kW) @30 °C ^[1]	5360		
	AC Output Power (kVA/kW) @35 °C ^[1]	5150		
	AC Output Power (kVA/kW) @40 °C ^[1]	4940		
	AC Output Power (kVA/kW) @50 °C ^[1]	4520		
	Operating Grid Voltage (kV)	34.5 kV ±10%	33 kV ±10%	30 kV ±10%
	Rated Impulse Withstand Voltage	145 kV		
	Operating Grid Frequency (Hz)	60 Hz	50 Hz	50 Hz
	Current Harmonic Distortion (THDi)	<3% per IEEE 519		
	Power Factor (CosPhi) ^[2]	0.5 leading ... 0.5 lagging		
	Reactive Power Compensation	Four quadrant operation		
	Overload Capability	166% - 100 ms / 150% - 5 s / 120% - 8 s / 110% - 15 s		
DC	DC Voltage Range Full Power ^[3]	1019 V – 1500 V		
	Maximum DC Voltage	1500 V		
	DC Voltage Ripple	<3%		
	Max. DC Continuous Current (A)	5367		
	Max. DC Short Circuit Current (kA)	500 kA with a time constant of 1 ms		
	Battery Technology	All type of batteries (BMS required)		
EFFICIENCY & AUX. SUPPLY	Efficiency (Max) (η)	98.00% including MV transformer		
	CEC (η)	97.53% including MV transformer		
CABINET	Dimensions [WxDxH] (ft)	21.3 x 6.5 x 7.5		
	Dimensions [WxDxH] (m)	6.5 x 2.0 x 2.3		
	Weight (lbs)	30865		
	Weight (kg)	14000		
	Type of Ventilation	Forced air cooling		
ENVIRONMENT	Degree of Protection	IP55		
	Operating Temperature Range ^[4]	From -40 °C to +60 °C, >30 °C power derating		
	Operating Relative Humidity Range	From 4% to 100%		
	Storage Temperature Range	From -40 °C to +60 °C		
	Max. Altitude (above sea level) ^[5]	2000 m	1000 m	
CONTROL	Communication Protocol	Modbus TCP		
	Grid Forming (VISMA) / Black Start Capable	Standard		
PROTECTIONS	Ground Fault Protection	Insulation monitoring device		
	Humidity Control	Active heating		
	General AC Protection & Disconnection	36 kV MV switchgear (2L+V)		
	General DC Protection & Disconnection	High-speed fuses, Motorized DC disconnect switches ^[6]		
	Overvoltage Protection	Type 2 for AC and Type 1+2 for DC		
STANDARDS	Safety	IEC 62109-1 / IEC 62109-2 / IEC 62477-1 / IEC 62477-2		
	Utility Interconnect ^[7]	IEC 62116 / / G99 / VDE 4110-4120-4130 / CEI 0-16 / NTS 2.1 / EN 50549		

NOTES

- [1] Values at 1.00·Vac nom and CosPhi=1. Consult Power Electronics for derating curves.
[2] Consult P-Q charts available: $Q(kVar) = \sqrt{(S(kVA))^2 - P(kW)^2}$.
[3] Consult Power Electronics for derating curves. In the event of overvoltage in the grid, the minimum DC voltage will vary proportionally with the AC voltage.
[4] Optional available for temperatures below -25 °C.
[5] Consult Power Electronics for altitudes above 1000 m.
[6] Battery short circuit disconnection must be done on the battery side.
[7] Consult Power Electronics for other applicable standards / grid codes.